A Synergistical Attention Model for Semantic Segmentation of Remote Sensing Images

Xin Li, Feng Xu, Fan Liu, *Member, IEEE*, Xin Lyu, Yao Tong, Zhennan Xu, and Jun Zhou, *Senior Member, IEEE*

***Abstract*— In remotely sensed images, high intraclass variance and interclass similarity are ubiquitous due to complex scenes and objects with multivariate features, making semantic segmentation a challenging task. Deep convolutional neural networks can solve this problem by modeling the context of features and improving their discriminability. However, current learning paradigms model the feature affinity in spatial dimension and channel dimension separately and then fuse them in a sequential or parallel manner, leading to suboptimal performance. In this study, we first analyze this problem practically and summarize it as attention bias that reduces the capability of network in distinguishing weak and discretely distributed objects from wide-range objects with internal connectivity, when modeled only in spatial or channel domain. To jointly model both spatial and channel affinity, we design a synergistic attention module (SAM), which allows for channelwise affinity extraction while preserving spatial details. In addition, we propose a synergistic attention perception neural network (SAPNet) for the semantic segmentation of remote sensing images. The hierarchical-embedded synergistic attention perception module aggregates SAM-refined features and decoded features. As a result, SAPNet enriches inference clues with desired spatial and channel details. Experiments on three benchmark datasets show that SAPNet is competitive in accuracy and adaptability compared with state-of-the-art methods. The experiments also validate the hypothesis of attention bias and the efficiency of SAM.**

***Index Terms*— Attention bias, contextual affinity, remote sensing images, semantic segmentation, synergistic attention.**

I. Introduction

SEMANTIC segmentation of remote sensing images aims to produce pixelwise categorical labels to facilitate interpretation of the remote sensing data [1], [2], [3]. The semantically parsed annotation enables an intuitive perception of targets and, therefore, has been widely adopted in downstream tasks, such as land-cover mapping [4], [5], water resources management [6], [7], and disaster assessment [8], [9], among others.

Due to the homogeneity between remote sensing and natural images, convolutional neural networks (CNNs) trained on large natural image datasets show strong adaptability and generalization capability when applied to remote sensing tasks [10], [11], [12], [13], [14]. Fully convolutional network (FCN) [15], for example, has become a baseline of semantic segmentation. The encoder–decoder architecture has also been adopted for this task. SegNet [16] gradually recovers the encoded abstract feature maps to the original spatial size to compensate for the sampling loss. U-Net [17] deploys multiple skip connections between encoder and decoder stages to retain details. Nevertheless, the abovementioned approaches suffer from the local receptive field that delivers information in a short-range context. Therefore, it is necessary to model sufficient and wider context to enrich inference clues based on CNNs.

To address this issue, dilated convolution was designed to extract multiscale features that help to obtain more accurate semantic information [18]. Dilated convolution, also named atrous convolution, paves a feasible way to adaptively modify the size of the field-of-view of the convolution filter. Consequently, the atrous convolution-based networks [19], [20] can aggregate a wider range of contextual information. Unfortunately, the global context cannot be well-extracted by stacking atrous convolutions [21], [22].

Attempting to enrich spectral–spatial context, Wang et al. [23] proposed to combine spectral, spatial, and semantic information before classification by a multikernel support vector machine, in which the spectral information, spatial information, and semantic information are processed by one kernel individually. Moreover, 3-D convolution is utilized. A 3-D fast learning block (depthwise separable convolution block and a fast convolution block) followed by a 2-D CNN was introduced to extract spectral–spatial features [24]. Tang et al. [25] developed a 3-D octave convolution with the spatial–spectral attention network to capture discriminative spatial–spectral features for the classification of hyperspectral images. Wang and Liang [26] developed a lightweight three-layer 3-D convolutional network module for hyperspectral images' spectral–spatial feature extraction. Alternatively, spectral–spatial fusion mechanism was investigated.

Manuscript received 30 September 2022; revised 15 January 2023; accepted 4 February 2023. Date of publication 10 February 2023; date of current version 23 February 2023. This work was supported in part by the Excellent Post-Doctoral Program of Jiangsu Province under Grant 2022ZB166; in part by the National Natural Science Foundation of China under Grant 42104033, Grant 82004498, Grant 51779100, and Grant 51679103; in part by the Project of Water Science and Technology of Jiangsu Province under Grant 2021080, Grant 2021063, and Grant 2021072; and in part by the Fundamental Research Funds for the Central Universities under Grant B210202080. *(Corresponding author: Feng Xu.)*

Xin Li, Feng Xu, Fan Liu, Xin Lyu, and Zhennan Xu are with the College of Computer and Information and the Key Laboratory of Water Big Data Technology of Ministry of Water Resources, Hohai University, Nanjing 211100, China (e-mail: li-xin@hhu.edu.cn; xufeng@hhu.edu.cn; fanliu@hhu.edu.cn; lvxin@hhu.edu.cn; zhennanxu@hhu.edu.cn).

Yao Tong is with the School of Artificial Intelligence and Information Technology, Nanjing University of Chinese Medicine, Nanjing 210023, China (e-mail: yaotong@uw.edu).

Jun Zhou is with the School of Information and Communication Technology, Griffith University, Nathan, QLD 4111, Australia (e-mail: jun.zhou@griffith.edu.au).

Digital Object Identifier 10.1109/TGRS.2023.3243954

1558-0644 © 2023 IEEE. Personal use is permitted, but republication/redistribution requires IEEE permission.
See https://www.ieee.org/publications/rights/index.html for more information.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 20:07:07 UTC from IEEE Xplore. Restrictions apply.

Li et al. [27] skillfully designed a gated mechanism to aggregate the context in spatial and spectral domains. Furthermore, Zhang [28] developed a two-stream spectral–spatial feature extraction network via 3-D hybrid convolution associated with a multilevel aggregation subnetwork. However, such kind of designs integrates multiple modules and always with massive parameters, hindering the availability and scalability of applying a segmentation network to remote sensing images with variable objects and complex scenarios.

Inspired by human perceptual and cognitive systems, the attention mechanism has advanced natural language processing and computer vision [29]. The attention mechanism allows networks to focus on salient regions or aspects of specific objects in complex scenes while filtering unimportant perceived information. In vision tasks, significant progress was made by SENet [30] that recalibrates channel weights based on their importance. This was followed by the proposal of self-attentiveness with a surprising ability to capture long-range dependencies. Nonlocal blocks [31] were initially designed for video classification and were successfully transferred to object detection and semantic segmentation. Similarly, DANet [32] implements the spatial attention module and channel attention module (CAM) in parallel and then conducts fusion operations to provide contextual information from both domains. Convolution block attention module (CBAM) [33] seamlessly integrates a lightweight attention module in a CNN to achieve competitive results in classification and detection. In addition, OCRNet [34] extracts relational context and uses it to sharpen the boundaries of objects in a self-attentive manner.

Remote sensing images acquired from high altitudes normally cover wide areas with complex and diverse backgrounds. The spatial details and potential semantic contents in remote sensing images are abundant even when the spatial resolution is relatively low [35], [36]. Thus, the ground objects are not well recognizable by most of computer vision networks. It is necessary to make an in-depth exploration of the visual features of objects in remote sensing images. Moreover, in different geographical areas, objects have typical variants. For example, forests have different colors and shapes in different areas and times, and Beijing National Stadium and Sapporo Dome have significant visual differences. Since CNNs cannot well-extract the latent inner class and interclass visual correlations of ground objects of remote sensing images, the attention mechanism provides an alternative to enlarge the interclass dissimilarities and reduce the inner class similarities. Driven by DANet [32], SCAttNet [37] adopts two lightweight attention modules that individually learn spatial and channel correlations, refining the learned representations adaptively. In addition, HMANet [38] exploits categorywise correlations beyond local-region attention to boost the discriminative capability of the segmentation model. Although achieving impressive results by diverting attention mechanism, the mutually independent scheme of modeling spatial and channel affinities is far from optimal.

No matter whether in a sequential or parallel fashion, the attention mechanism cannot model all relationships between spatial and channel domains. For example, in traditional channel attention, each channel is connected to other channels while ignoring spatial responses in a particular feature map. Similarly, interactions between channels are missing in the spatial attention procedure. The consequence of this drawback is a condensation of attentional features, named attentional bias. Specifically, TESA [39] captures interdependencies self-attentively along all dimensions of the tensor using a postfusion of vector similarities to compensate for such bias. Although the fused similarities are more accurate than individuals, the vector similarities initially extracted by TESA are independent, which inevitably leads to attentional bias. However, the integrity of multidimensional contextual information for understanding spatial and channel interactions is ubiquitous in the vanilla visual system. This inheritance allows humans to become intelligent and accurate in perceiving and understanding panoramas. Thus, a promising solution to alleviate attentional bias is necessary.

To sum up, there are two problems with the existing attention-based semantic segmentation networks for remote sensing images.

1) Different classes of objects in remote sensing images may share similar color or texture properties. Objects of the same class may have various spatial sizes or shapes. Both textural information and spatial information are necessary for accurate ground object recognition in remote sensing images. Deploying spatial and CAMs sequentially or parallelly leads to the one sidedness of their contextual information, regardless of the completeness of the context across spatial and channel domains. Eventually, this design might potentially result in suboptimal segments of weak and small objects or inconsistency inside uniformly represented objects.
2) From the computational complexity perspective, calculating the dependencies of spatial and channel domains separately leads to higher memory and computation costs as it may involve massive matrix multiplications. Therefore, it is a high priority to reduce matrix manipulations while retaining sufficient affinity extraction and aggregation, in which the spatial context is considered while generating channelwise dependencies.

To solve these two problems, in this article, a synergistic attention module (SAM), which collaboratively models spatialwise and channelwise correlations into one unique attention map, is devised to capture contextual affinity across spatial and channel domains simultaneously. This module allows the network to leverage a more comprehensive affinity that both spatial and channel correlations involved to refine representations. Essentially, we unify the attention for both dimensions in an individual model that accepts spatial affinity when computing channelwise attention maps. Furthermore, both channelwise and spatialwise interactions are jointly modeled as attentional similarities to complement inference clues. Finally, we implement plug-and-play SAM to form a synergistic attention perception neural network (SAPNet) for semantic segmentation. Experiments are conducted to validate the efficiency and superiority on three remote sensing benchmarks. The threefold contributions of this article are summarized as follows.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 20:07:07 UTC from IEEE Xplore. Restrictions apply.

1) We undertake a theoretical and experimental analysis to prove the attention bias and its impact on the semantic segmentation of remote sensing images. This leads to the proposal and design of a consolidated attention module to address attention bias and retain integrated feature representations. SAM models well-preserved context and spatial and channel dependencies with the rationale of multidimensional nonlocal attention. Moreover, the proposed SAM requires fewer matrix manipulations, reducing the computational complexity and execution time.
2) A novel neural network called SAPNet is proposed. SAPNet hierarchically deploys SAMs between the encoder and the decoder. Thus, the contextual information is more comprehensive without massive spatial or channel loss. Consequently, the uncertainty of classifying one pixel into easy-confused classes is significantly reduced.
3) Extensive experiments are conducted on three representative remote sensing image semantic segmentation benchmarks [40], [41], [42]. The results show that SAPNet is superior to several state-of-the-art methods. Furthermore, the efficacy and efficiency of SAM are validated and analyzed.

The rest of this article is organized as follows. Section II presents the related works about semantic segmentation of remote sensing images and affinity modeling. Section III introduces the proposed method in detail. Section IV contains experiments, comparisons, and discussions to validate SAPNet on three remote sensing benchmarks of semantic segmentation. Finally, Section V draws the conclusions.

II. Related Works

A. Affinity Modeling

Contextual affinity is used to refine learned representations that are typically generated by the convolutional backbones [43], [44]. Early affinity modeling approaches were realized by deploying dilated convolutions. DeepLab [18] designs and utilizes pyramid dilated convolutions with multiple rates to enlarge receptive fields. Thus, such convolution allows the feature to contain a wider range of perceptual information than the neighboring pixels. On this basis, DenseASPP [45] densifies the dilated rates to cover more extensive scale ranges. However, due to the discontinuity of the convolution kernels at different scales, the dilated convolution leads to the loss of local features. In addition, the captured long-range context may be irrelevant.

Recently, the graph convolutional network (GCN) has been adopted to describe non-Euclidean data that lack shift invariance. For semantic segmentation, Bronstein et al. [46] proposed a dual graph convolutional network (DGCNet) that models the global context of the input feature by modeling two orthogonal graphs in a single framework. However, the convolutional feature map is necessary in DGCNet and only a slight improvement has been obtained on segmenting natural images. Moreover, Zhang [47] developed a CNN-enhanced heterogeneous graph convolutional network (CHeGCN) for park segmentation, constructing the relational topology of a binary classification task. Nevertheless, the construction of relation graph of multiclass task would be very complex due to the inherence of ground objects in remote sensing images, hindering the extension of GCN-based methods.

To selectively capture beneficial long-range context, the attention mechanism has been proposed. On the one hand, the attention mechanism determines whether correlations between input elements should be considered or not. On the other hand, the attention mechanism quantifies how much weight should be put into it. To date, the attention mechanism has been widely used in remote sensing, including scene classification [48], [49], object detection [50], [51], pansharpening [52], [53], and change detection [54], [55]. The essence of the attention-based deep neural network is to incorporate the similarity between different channels or positions to enhance pixel representations. As a result, the enhanced representations are more discriminative. Thus, the predictive results usually have higher accuracy.

Early attention mechanisms for vision tasks focus on modeling affinity in either channel or spatial dimension. For example, SENet [30] first highlighted the informative feature channels and weaken the less useful ones by adaptively recalibrating the channels with specific weights in vision tasks. Concurrently, Wang et al. [31] introduced nonlocal operators to capture spatialwise correlations to enrich spatial details, which inaugurated the exploration of the self-attention mechanism. However, the 1-D contextual affinity is insufficient to produce discriminative features. Instead of identifying features only in the spatial domain, CBAM [33] attempted to emphasize the informative features in both spatial and channel domains by stacking channel attention and spatial attention. As a semantic segmentation-specific attention network, DANet [32] adopts the self-attention mechanism instead of stacking convolutions. More concretely, DANet deploys the position attention module (PAM) and CAM in parallel to capture dependencies. Latterly, transformer and its variants based on self-attention mechanism have also been proposed for semantic segmentation. However, the initial tokenization at the beginning of transformer would break the inherent details of ground objects, such as structure or boundary. For dense predictions, such details are essential for localization and recognition [56].

To sum up, nonlocal techniques provide a simple yet effective way to adaptively capturing position/pixelwise contextual information. The nonlocal-based module can be flexibly embedded in different stages of CNN backbone with feature maps at different sizes, thus enabling multiscale feature refinement and context enrichment.

Although exceptional results have been achieved by introducing attention mechanisms, the sequential or parallel deployment of attention modules is individually biased. The generated attention maps miss the information from the counterpart dimension. Unlike existing attention methods, we propose to synergistically model the contextual affinity across spatial and channel dimensions. Apart from recently used transformers, this one-stage attention module allows joint and simultaneous modeling of the contextual affinity in a unified attention map.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 20:07:07 UTC from IEEE Xplore. Restrictions apply.

B. Semantic Segmentation of Remote Sensing Images

Remote sensing images are normally acquired at high altitudes. They often cover diverse and complex ground objects with high intraclass variance and interclass similarity. Semantic segmentation labels every pixel for the delineation of objects. Methods are typically implemented with statistical analysis and handcrafted feature-aided learning or data-driven deep learning steps.

Some state-of-the-art remote sensing semantic segmentation approaches adopt the encoder–decoder architecture due to its stable and reliable feature extraction and transformation capability. To boost localization accuracy and maintain feature consistency, U-Net is extensively analyzed and explored for remote sensing imagery. TreeUNet [57] adaptively increases the classification accuracy at the pixel level by building a Tree-CNN block. Tree-CNN block is constructed with a confusion matrix to fuse multiscale features. ResUNet-a [58] extends the U-Net baseline by introducing residual connections, atrous convolutions, pyramid scene parsing, pooling, and multitasking inference. To our knowledge, this network has massive parameters while achieving excellent accuracy.

Recently, attention-based deep neural networks have been introduced for the semantic segmentation of remote sensing images. For example, Li et al. [59] proposed to use a deep layer CAM and a shallow layer spatial attention module to segment large-scale satellite imagery. Moreover, the squeeze and excitation (SE) design is extended to spatial dimension to focus on patchwise semantics, bridging the semantic gap between high- and low-level features [60]. AFNet [61] designs a multipath encoder structure, a multipath attention-fused module, and a refinement attention-fused block for semantic segmentation of very-high-resolution remote sensing images. HMANet [38] hybridizes three-level representations to augment learned features, including space, channel, and category. Similarly, Li et al. [62] proposed HCANet that extracts and aggregates cross-level contextual information from pixels, superpixels, and global space. In addition, an adaptive fusion network was designed by Liu et al. [63], enhancing the feature points from either low- or high-level feature maps with the constructed attention modules. LANet introduces a patchwise attention module to incorporate locality information and then adaptively fuses cross-level attention via devised attention embedding module [64].

In the context of semantic segmentation of remote sensing images, spatial details and channel information are both important to highlight the difference between diverse objects. Although the incorporation of the attention mechanism enables the segmentation network to capture correlations in all dimensions, the attention module inevitably misses the intrinsic spatial–channel interactions, leading to attention bias. To capture these interactions, this study strives to synergistically model multidimensional dependencies in a single module. This module generates a single map instead of aggregating multiple contextual affinity maps.

III. Method

First, an overview of the proposed network is presented in Section III-A. Next, the attention bias is discussed based on theoretical and practical analysis in Section III-B. Then, Section III-C thoroughly illustrates the proposed SAM. A complexity analysis of SAM is also described in Section III-D. Finally, the pipeline of synergistic attention perception (SAP) and framework of SAPNet is presented in Section III-E.

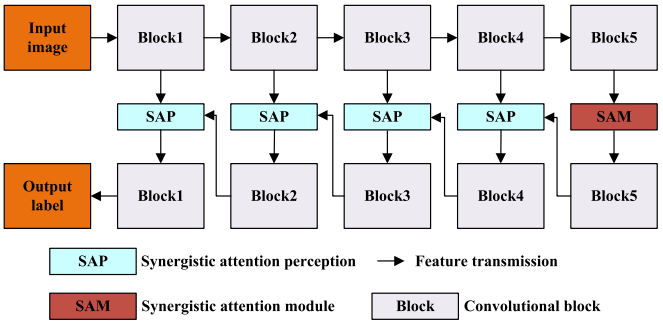


Fig. 1. Overview of the proposed method.

A. Overview

Fig. 1 shows a flowchart of the proposed method. The designed network based on the encoder–decoder architecture deploys one SAM and four SAP modules between the encoder and decoder blocks. SAM refines the input features with spatial and channel dimensional affinities. Also, SAP allows aggregation of encoded, decoded, and SAM-refined features to preserve informative details. Therefore, the discriminability of pixelwise representations is enhanced before prediction. The details are given in the following.

B. Attention Bias

Contextual affinity plays a pivotal role in augmenting representations, highlighting valuable details, and disregarding irrelevant parts. Therefore, it is essential to introduce rich details into learned representations for pixel-level dense prediction [65]. Due to the advancements in attention mechanisms, the primary current focus in affinity modeling is on how to explore attention modules efficiently and effectively. The nonlocal rationale is generalized in spatial and channel dimensions among all the attention variants. To comprehensively demonstrate the attention bias, we take DANet [32] as an example. DANet has a PAM for spatial contextual affinity and a CAM for channelwise contextual affinity, denoted as PAM and CAM, respectively.

Fig. 2 shows the pipelines of PAM and CAM, and SAM is shown in Fig. 5 (top). Given an input feature $F_{\text{in}} \in \mathbb{R}^{H\times W\times C}$, where H, W, and C denote height, width, and channel, respectively, in PAM, the vanilla nonlocal block is adopted. Apart from PAM, CAM feeds features into three branches without convolution for preserving the original channel correlations. The contextual affinity of PAM and CAM is generated with a spatial size of $HW \times HW$ and $C \times C$, respectively. From the perspective of feature transformation, spatial affinity is the intradimensional correlation between arbitrary position pairs, ignoring the impacts from other channels even at the same position. As to CAM, each channel can be connected with other channels while discounting spatial interactions.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 20:07:07 UTC from IEEE Xplore. Restrictions apply.

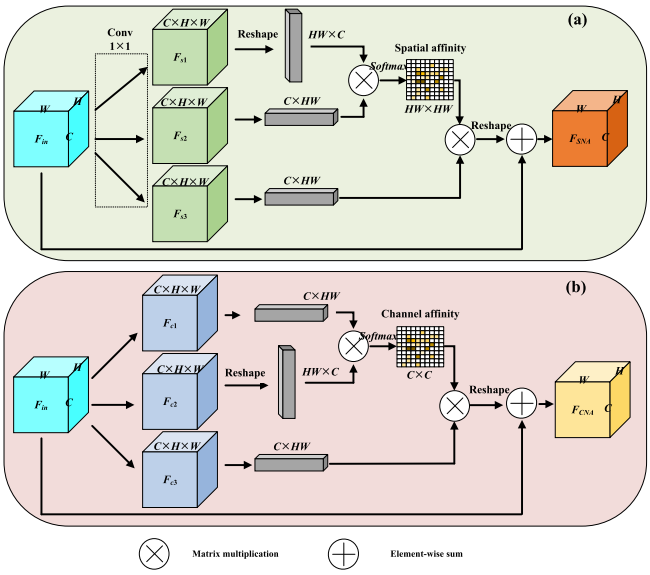


Fig. 2. Pipelines of (a) PAM and (b) CAM.

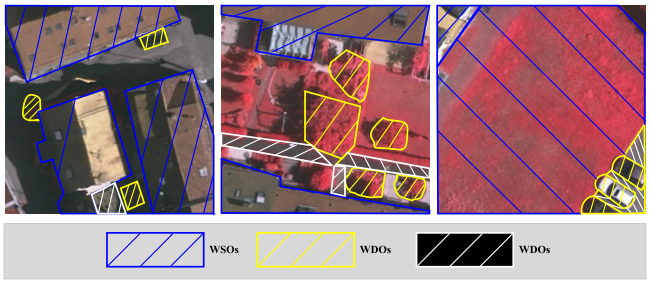


Fig. 3. Examples of Vaihingen for understanding WDOs and WSOs.

Remote sensing images usually cover a wide range of areas with high intraclass variance and low interclass differences. As a result, weak and small objects are often misclassified. Moreover, the consistency inside the big object cannot be guaranteed. To address these issues, we propose an SAM that releases attention bias by synergistically modeling contextual affinity in a single attention map.

We further conduct an exploratory experiment on the ISPRS Vaihingen benchmark to verify the attention bias issue. The predefined objects in the remote sensing image can be divided into two categories: weak and discrete-distributed objects (WDOs), and wide-range with strong internal connectivity objects (WSOs). As shown in Fig. 3, three subpatches of Vaihingen are used to visually understand WDOs and WSOs. The WSOs are delineated with blue lines, while WDOs are with yellow or white lines.

The experimental models are shown in Fig. 4. The SegNet with ResNet 50 is adopted as the backbone for the baseline. At the end of the encoder, five different modules are embedded for feature refinement. In addition to deploying a single attention module, such as SAM, PAM, and CAM, the sequential and parallel fusion models are also examined. Besides, we test the attention bias based on the proposed SAPNet with these attention models embedded. The comprehensive experimental results are presented in Section IV-D.

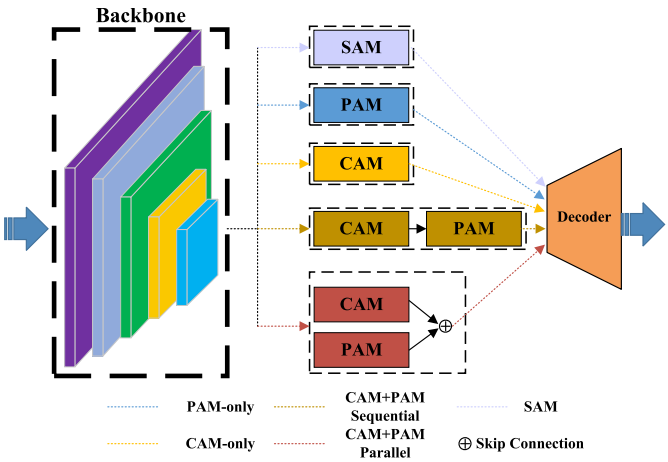


Fig. 4. Schemes of exploratory experiments with five models.

In general, modeling spatial or channel contextual affinity independently leads to attentional bias. Extracting channel attention ignores the spatial correlations inside the specific channels. As a result, the refined features might not be apparently discriminative. Therefore, we suggest that affinities shall be modeled synergistically to represent both positional and channel information in a unique attention map.

C. Synergistic Attention Module

As previously discussed, existing nonlocal attention yields attention bias to specific dimensions while ignoring others. Moreover, the computations are redundant with a sequential or parallel fusion model. To model the contextual affinity in a sole attention map without sacrificing computational efficiency, an SAM is devised.

The core pipeline is shown in Fig. 5, in which the bottom subplot shows the structure of feature transformation and reorganization modules (F&R). The input feature map is denoted as $F_{\text{in}} \in \mathbb{R}^{C\times H\times W}$, where H, W, and C represent width, height, and channel, respectively. First, F&R is deployed to extract contextual priors over horizontal and vertical coordinates in the spatial dimension. The widthwise and heightwise global average pooling are designed to obtain richer context horizontally and vertically, ensuring spatial consistency when computing channelwise affinity. Then, $\text{Pool}(F_{\text{in}})_W \in \mathbb{R}^{C\times 1\times W}$ and $\text{Pool}(F_{\text{in}})_H \in \mathbb{R}^{C\times H\times 1}$ are generated. Using different sizes of pooling with regard to height and width enables the extraction of global contextual information along both horizontal and vertical directions. After H and W times of loops, reconstructed features $F_w \in \mathbb{R}^{C\times H\times W}$ and $F_h \in \mathbb{R}^{C\times H\times W}$ are obtained. Afterward, the reconstructed features are split along the corresponding axial direction, followed by concatenation. Therefore, the output of F&R is $F_{\text{fr}} \in \mathbb{R}^{C\times S\times (H+W)}$, where $S = H = W$ in the experiments.

F&R outputs the refined feature map that introduces spatial contextual as priors. Subsequently, the transposed F_{fr}^T is left multiplied by F_{fr} to calculate contextual affinity

$$A(i,j) = \frac{\exp\left(F_{\text{fr}}(i) \bullet F_{\text{fr}}^T(j)\right)}{\sum_{i=1}^{C} \exp\left(F_{\text{fr}}(i) \bullet F_{\text{fr}}^T(j)\right)} \tag{1}$$

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 20:07:07 UTC from IEEE Xplore. Restrictions apply.

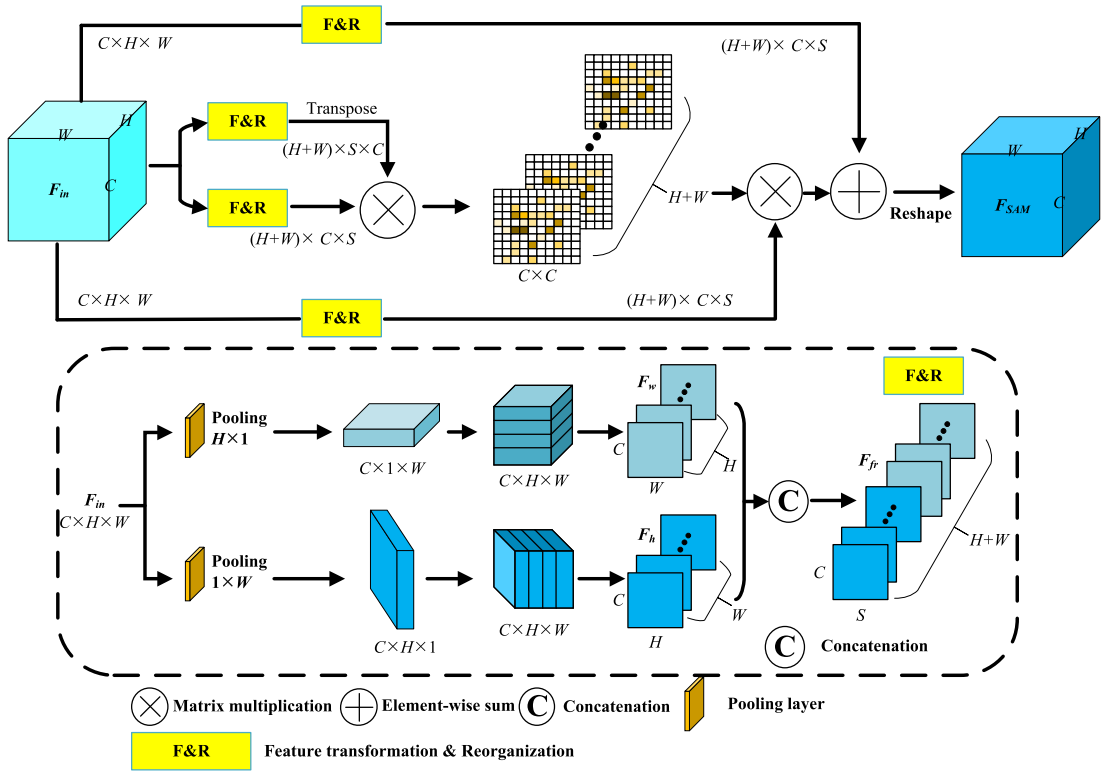


Fig. 5. Detailed structure of SAM.

where $A(i, j) \in \mathbb{R}^{(H+W)\times C\times C}$ is the correlation between the ith and jth channels at a specific spatial position.

Finally, matrix multiplication and an elementwise summation are implemented, injecting a 3-D contextual affinity in which the spatial context served as the prior in calculating channelwise correlation. Thus, it can be concluded that both spatial and channel contexts are aggregated. This step is implemented as follows:

$$F_{\mathrm{SAM}}(j) = \alpha \sum_{i=1}^{C} A(i, j) \bullet F_{\mathrm{fr}}(i) + F_{\mathrm{fr}}(j) \tag{2}$$

where α is a learnable coefficient, $F_{\mathrm{SAM}} \in \mathbb{R}^{C\times H\times W}$ is the refined representations with spatial and channel contextual affinities, and $F_{\mathrm{SAM}}(j)$ is the jth channel of F_{SAM}.

Here, the contextual affinity captured by SAM does not compress the spatial information while capturing the channel-dependent attention maps. SAM assumes that spatial correlations can be treated as contextual priors and injected into the channelwise attention calculation. In this way, the SAM reaches a holistic contextual view, boosting the discriminability of learned representations. This property makes the network robust to diverse and complex scenarios of remote sensing images.

D. Complexity of SAM

Given an input feature with a size of $C \times H \times W$, the computational complexity of PAM is $O((HW)^2C)$, while CAM has a complexity of $O(C^2HW)$. PAM and CAM model the dependencies of spatial and channel dimensions individually. To model 3-D dependencies, sequential and parallel architectures separately calculate spatial and channelwise correlations, yielding higher complexity and requiring more GPU memory. SAM leads to a novel way of computation by jointly modeling spatial and channel dimensional correlations, which has a computational complexity of $O(C^2(H + W)S)$, where $S = H = W$ in the experiments. Therefore, SAM has the same order as CAM while introducing a small number of matrix multiplications.

E. Synergistic Attention Perception Neural Network

As shown in Fig. 6, the schematic of the proposed SAPNet is presented. The network consists of an encoder–decoder baseline, four SAP modules, and one SAM. The upper branch is the feature encoding path, while the bottom is the decoding path. Four SAP modules are designed and embedded hierarchically for refining features with different scales. Moreover, the encoded features at the most miniature scale have a very small spatial size where limited spatial details are available. Furthermore, there is no scale variation in transforming this feature to the decoder stage. Therefore, the output features of the encoder are refined with a single SAM.

The encoded feature is given as $F_e(l), 1 \le l \le 5$, which represents the output feature of specific encoder stages. Correspondingly, $F_d(l), 1 \le l \le 5$, denotes the output of different decoder blocks. Specifically, $F_d(5)$ can be formed as

$$F_d(5) = F_{\mathrm{SAM}}(F_e(5)) \tag{3}$$

where $F_{\mathrm{SAM}}(\cdot)$ refines $F_e(5)$. The SAP module is utilized to aggregate the encoded and decoded features to prevent

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 20:07:07 UTC from IEEE Xplore. Restrictions apply.

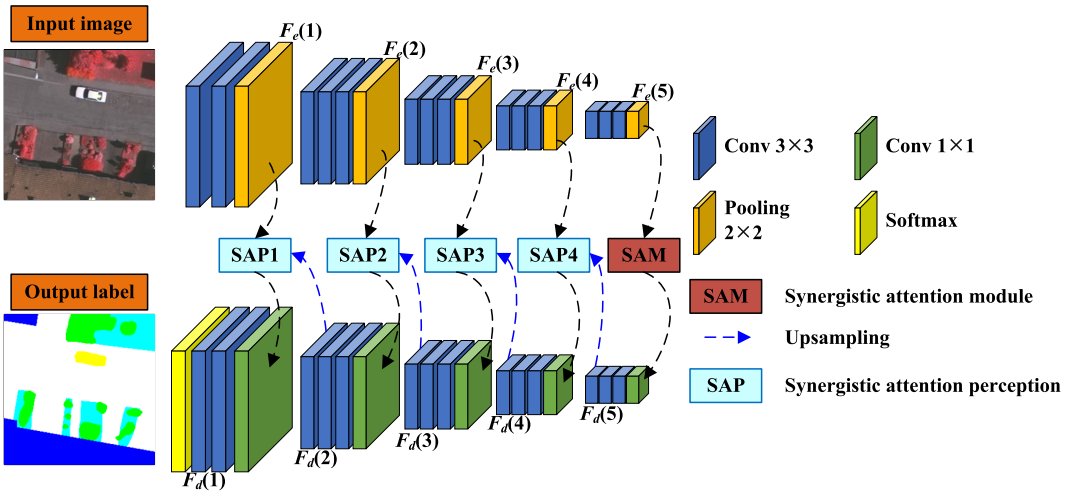


Fig. 6. Framework of SAPNet.

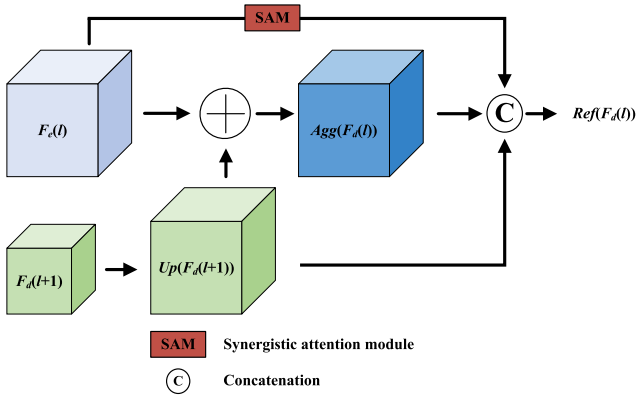


Fig. 7. Pipeline of SAP module.

information loss, in which SAM is deployed as a branch for feature refinement.

Fig. 7 shows the SAP module. SAP initially aggregates the upsampled features from the decoder with encoded ones. Then top branch outputs the features on contextual affinity by SAM. Finally, concatenation followed by a 1×1 convolution is implemented for generating sophisticated features as follows:

$$\mathrm{Agg}(F_d(l)) = (F_e(l) \oplus Up(F_d(l+1))) \tag{4}$$

$$\mathrm{Ref}(F_d(l)) = \mathrm{Conv}_{1\times1}\left(\mathrm{Concat}\begin{pmatrix} F_{\mathrm{SAM}}(F_e(l)) \\ \mathrm{Agg}(F_d(l)) \\ \mathrm{Up}(F_d(l+1)) \end{pmatrix}\right) \tag{5}$$

where $\mathrm{Ref}(F_d(l))$ represents the input feature for decoder l that derives from the fusion of the features from the former decoder and counterpart encoder, Up(.) denotes the upsampling operator, and Concat(.) denotes the concatenation operator.

Overall, the encoder–decoder baseline is mutated with multiscale feature refinement, in which all-inclusive contextual affinity and well-preserved original details are merged. As a result, comprehensive clues are provided for the probabilistic delimitation of each pixel.

IV. Experiments and Discussion

We have conducted extensive experiments on three benchmarks to compare the performance of the proposed method with several state-of-the-art methods. We also evaluate attention bias with four attention models based on two baselines.

A. Datasets

1) ISPRS Vaihingen Dataset: It was acquired from an aerial platform. Its spatial resolution is 9 cm. Six land-cover classes were labeled at the pixel level. Specifically, clutter serves as a background class and is ignored in accuracy evaluation. The publicly available data consist of 33 true orthophoto tiles with an average spatial size of 2494 × 2064 pixels. Red, green, and near-infrared (NIR) bands form the false-color images. The numbers of training, validation, and testing images are 14, 2, and 17, respectively.

2) ISPRS Potsdam Dataset: It was also acquired from an aerial platform but with a spatial resolution of 5 cm. The same land-cover classes as Vaihingen were labeled at the pixel level. Four bands, R, G, blue (B), and NIR, are used. The spatial size of each image is 6000 × 6000. There are 26 images for training, four images for validation, and eight images for testing.

3) DeepGlobe Land-Cover Classification Dataset: It was acquired from the satellite with a spatial resolution of 0.5 m and three bands (R, G, and B). Compared with aerial images, satellite imagery suffers more from the diversity of land covers. The dataset contains 1146 images with a spatial size of 2448 × 2448 pixels. Among them, 802 images are used for training, 114 images are used for validation, and 230 images are used for testing.

B. Implement Details

The experiments were implemented using PyTorch with an NVIDIA 3090 GPU under Linux OS. All images with ground truth were cropped with a spatial size of 256 × 256 for training, validation, and testing. The setting and hyperparameters are listed in Table I. Moreover, standard scaling (0.5, 1.5, and 2.0), rotation (90°, 180°, and 270°), and horizontal and vertical flips were employed for data augmentation.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 20:07:07 UTC from IEEE Xplore. Restrictions apply.

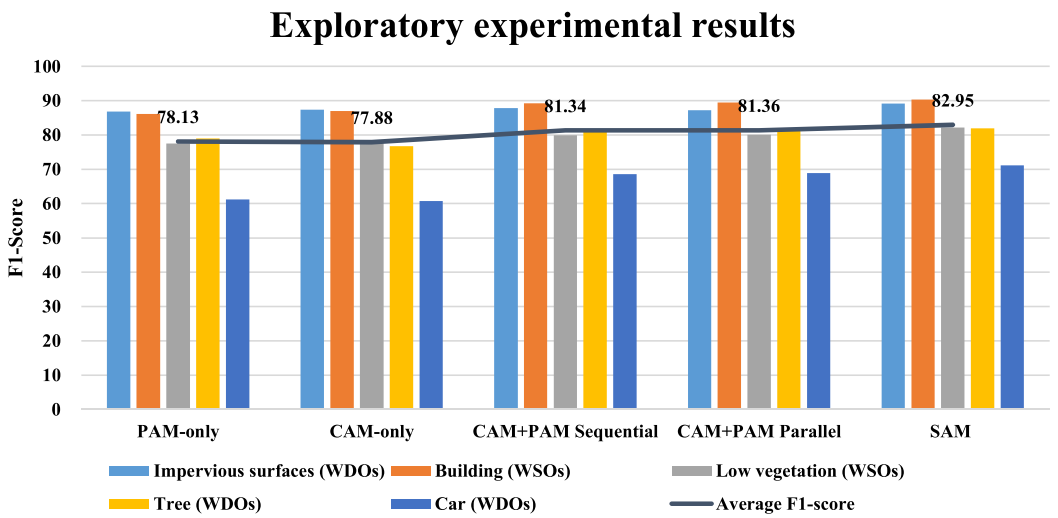


Fig. 8. Exploratory experimental results of different attention models based on SegNet, including categorywise F1-score and AF.

TABLE I
HYPERPARAMETER SETTINGS

Items	Settings
Backbone	ResNet 101
Learning strategy	Poly decay
Initial learning rate	0.005
Loss function	Softmax cross-entropy
Optimizer	Adam
Max epoch	500
GPU memory	24 GB
Sub-patch size	256×256
Batch size	16

We applied softmax cross entropy as the loss function for all methods in experiments.

To make a convincing comparison, we reproduced and evaluated ten methods with the same settings as in their original papers, except for the batch size and the optimizer. Among these methods, FCN-8s [15], SegNet [16], U-Net [17], and DeepLab V3+ [18] were pioneer encoding and decoding networks. CBAM [33] and DANet [32] were selected as attention-based natural image segmentation networks. RAANet [66], MACU-Net [67], SCAttNet [37], LANet [64], and HMANet [38] are the state-of-the-art semantic segmentation methods developed for remote sensing images specifically. In addition, a hybrid network ResUNet-a [58] was included as a nonattentive segmentation network for remote sensing images. It is worth noting that the reported results may not be identical to those in the original paper. Among them, the reproduced results of RAANet, MACU-Net, SCAttNet, LANet, HMANet, and ResUNet-a are slightly lower in accuracy than those published in the original paper. We also used a random approach for data partitioning, so the partition may not be exactly the same as the others either.

C. Evaluation Metrics

Similar to SCAttNet [37] and LANet [64], we adopted the overall accuracy (OA) and average F1-score (AF) to evaluate the performance of all methods. The OA is defined as the ratio of correctly classified pixels to all pixels.

For classwise evaluation, the F1-score is calculated according to precision and recall

$$F_1 = 2 \cdot \frac{\text{precision} \cdot \text{recall}}{\text{precision} + \text{recall}} \tag{6}$$

where precision and recall are calculated as

$$\text{precision} = \frac{\text{TP}}{\text{TP} + \text{FP}} \tag{7}$$

$$\text{recall} = \frac{\text{TP}}{\text{TP} + \text{FN}} \tag{8}$$

where TP, FP, and FN are the numbers of true positive, false positive, and false negative samples, respectively. Furthermore, the AF of all classes is also calculated.

D. Exploratory Experiment for Evaluating Attention Bias

1) Embedding Different Attention Models Into SegNet: The exploratory experiments were conducted on the ISPRS Vaihingen benchmark to validate the attention bias hypothesis described in Section III-B. We reimplemented four models. The details of these models are shown in Fig. 4. PAM-only denotes the network that deploys PAM after the encoder to refine representation. CAM-only is similar to PAM-only but replaces PAM with CAM. CAM + PAM sequential embeds PAM followed by CAM sequentially. CAM + PAM parallel organizes the PAM and CAM in parallel and then fuses them with an elementwise summation. Note that the proposed SAM is also embedded in the same baseline. The dotted lines with different colors represent the corresponding flows of models in the experiments. The standard SegNet [16] with ResNet50 as the backbone forms the baseline. The hyperparameter settings and data partition are the same for each model for fairness. The classwise F1-score and AF on the test set for each method are shown in Fig. 8.

It can be seen that PAM-only and CAM-only produce a lower accuracy than combined ones. However, in addition to

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 20:07:07 UTC from IEEE Xplore. Restrictions apply.

TABLE II
NUMERICAL RESULTS ON THE ISPRS VAIHINGEN DATASET OF DIFFERENT ATTENTION MODELS BASED ON SAPNET. CATEGORYWISE F1-SCORE, AF, AND OA ARE LISTED, WHERE BOLD INDICATES THE BEST

Methods	Impervious surfaces	Building	Low vegetation	Tree	Car	AF	OA
PAM-only	81.7	89.2	70.9	88.2	71.6	80.32	84.6
CAM-only	86.8	91.8	74.4	84.6	63.2	80.16	83.1
CAM+PAM sequential	87.8	92.0	78.4	86.2	72.7	83.42	85.9
CAM+PAM parallel	88.2	92.4	79.8	86.7	75.7	84.56	86.5
SAP (with proposed SAM)	**94.2**	**96.2**	**85.6**	**91.6**	**91.7**	**91.86**	**89.7**

almost the same level of accuracy, PAM-only outperforms on WDOs, including impervious surfaces, trees, and cars, while CAM-only performs better on WSOs, including buildings and low vegetation. The combined models show significant improvement in performance. The OA of CAM + PAM sequential and CAM + PAM parallel increases by more than 3% over the PAM-only or CAM-only models. This observation verifies that both spatial and channel affinities are essential for feature refinement. However, the integration of two attention-refined features is implemented as a postprocessing step, introducing attention bias. Our proposed SAM synergistically models the contextual affinity across channels and spatial within one attention map, enriching the inference clues. The exploratory experiment with standard SegNet as a baseline shows a remarkable improvement of the proposed method with about 2% higher than CAM + PAM sequential and CAM + PAM parallel.

2) Embedding Different Attention Models Into SAPNet: We further examined the attention bias based on the proposed SAPNet (see Fig. 6) on the ISPRS Vaihingen benchmark. Four models were implemented with different attention modules, namely, PAM-only, CAM-only, CAM + PAM sequential, and CAM + PAM parallel (see Fig. 4). The numerical results are listed in Table II and the training losses are presented in Fig. 9.

Table II reveals that there has been a steady increase in the values of AF and OA with the proper integration of PAM and CAM models. Notably, the proposed SAP model correctly classified 89.7% of predicted pixels, achieving a large margin of 3.2% over the CAM + PAM parallel model. Superior performance is also observed in the value of AF. As for classwise analysis, we can see that the OA of cars has experienced an increase of more than 20% than the single-dimensional context. Regarding the buildings, strong inside-consistent distribution requires the assistance of a 3-D context, which makes SAP achieve the best F1-score of 96.2.

A convincing comparison is presented between PAM-only and CAM-only models, authenticating the hypothesis of attention bias in segmenting remote sensing images. As predefined, the impervious surfaces, buildings, and low vegetation belong to WSOs. When calculating channel attention maps, the global average pooling causes less loss. As a result, the F1 scores of these three categories of CAM-only are higher than PAM-only. As for WDOs, the global average pooling of CAM-only leads to the loss of spatial details, dropping the F1-score with 3.6% of trees and 8.4% of cars with respect to PAM-only. PAM concerns the spatial correlations that are essential for WDOs.

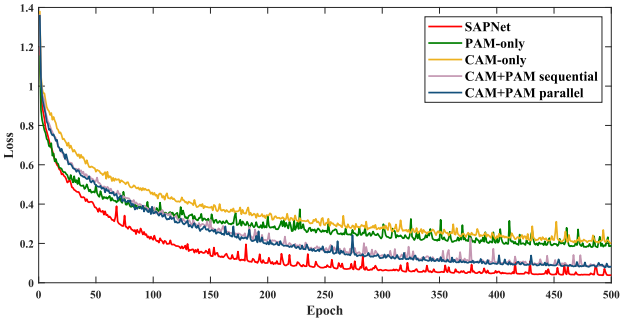


Fig. 9. Training loss of different attention models based on SAPNet.

CAM + PAM sequential and CAM + PAM parallel exhibit successive increases. The AF and OA results suggest that parallel fashion is better than cascaded ones for remote sensing images. Optimally, the unified manner of SAM, embedded in SAP, captures a 3-D context, leading to significant accuracy improvement.

We apply commonly used softmax cross entropy as the loss function for all comparative models. As shown in Fig. 9, the training loss is demonstrated. Less loss means higher accuracy. Compared to other models, SAP significantly reduces the loss with steady convergence at a comparatively fast speed. It is evident that the training loss is in exceptionally good agreement with numerical results.

To sum up, SAM can capture both spatial and channel contexts to boost pixelwise representations, avoiding information loss in learning. Thus, the provided contextual information is more beneficial for inference.

E. Comparison With the State-of-the-Art Methods

1) Results on the ISPRS Vaihingen Dataset: Table III quantitatively compares the results of all methods on the ISPRS Vaihingen dataset. It is evident that AF and OA strongly favor the performance of SAPNet, with AF and OA being 91.26 and 89.1%, respectively. Compared to natural image-specific methods, including FCN-8s, SegNet, U-Net, and DeepLab V3+, SAPNet significantly increases the performance by at least 13.92 AF and 13.6% OA. The CBAM and DANet introduce attention modules. Although their performance improvement is not significant, they confirm that the utilization of context paves a promising way to refine learned representations. RAANet embeds CBAM into DeepLab V3+ to enrich the contextual information, raising 0.3% of OA compared with DeepLab V3+. Based on U-Net, MACU-Net further realigns semantic features by multiscale skip

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 20:07:07 UTC from IEEE Xplore. Restrictions apply.

TABLE III
RESULTS ON THE ISPRS VAIHINGEN DATASET. CLASSWISE F1-SCORE, AF, AND OA ARE LISTED, WHERE BOLD TEXT INDICATES THE BEST RESULTS

Methods	Impervious surfaces	Building	Low vegetation	Tree	Car	AF	OA
FCN-8s [15]	85.9	75.2	66.5	81.7	39.6	69.78	68.1
SegNet [16]	89.1	77.0	69.2	84.6	43.5	72.68	70.9
U-Net [17]	86.5	88.0	71.0	84.5	42.8	74.56	72.8
DeepLabV3+ [18]	89.9	89.7	74.3	87.4	48.4	77.94	76.1
CBAM [33]	91.9	87.5	78.4	89.0	57.3	80.82	78.9
DANet [32]	91.1	89.9	79.6	87.7	58.2	81.30	79.4
ResUNet-a [58]	92.5	95.1	85.1	88.9	91.4	90.60	88.4
RAANet [66]	90.9	89.7	76.2	86.1	48.5	78.28	76.4
MACU-Net [67]	89.5	90.1	74.3	87.0	62.7	80.72	78.8
SCAttNet [37]	89.1	90.3	80.0	80.3	70.5	82.06	80.1
LANet [64]	93.8	95.7	82.9	**91.9**	88.4	90.54	88.4
HMANet [38]	92.7	96.1	84.8	89.9	87.9	90.28	88.1
SAPNet (ours)	**94.2**	**96.2**	**85.6**	91.6	**91.7**	**91.86**	**89.7**

TABLE IV
RESULTS ON THE ISPRS POTSDAM DATASET. CLASSWISE F1-SCORE, AF, AND OA ARE LISTED, WHERE THE BOLD TEXT INDICATES THE BEST RESULTS

Methods	Impervious surfaces	Building	Low vegetation	Tree	Car	AF	OA
FCN-8s [15]	86.0	75.3	66.6	81.8	39.7	69.90	68.8
SegNet [16]	86.8	87.0	71.2	84.8	42.3	74.40	73.3
U-Net [17]	90.1	89.9	74.4	87.5	48.5	78.07	76.9
DeepLabV3+ [18]	88.7	91.5	82.2	82.2	90.8	87.08	85.7
CBAM [33]	90.2	95.2	84.1	85.0	86.1	88.12	87.8
DANet [32]	90.6	95.9	83.5	83.3	93.1	89.28	87.9
ResUNet-a [58]	93.4	96.8	87.6	88.3	**96.1**	92.44	91.0
RAANet [66]	90.4	95.7	87.2	81.8	77.6	86.54	85.2
MACU-Net [67]	89.2	92.8	86.8	83.0	79.2	86.20	85.9
SCAttNet [37]	91.4	96.9	84.8	86.6	92.3	90.40	88.6
LANet [64]	92.4	96.4	86.8	87.7	93.4	91.34	90.2
HMANet [38]	93.3	97.0	87.7	88.4	96.2	92.52	91.1
SAPNet (ours)	**94.4**	**97.7**	**87.8**	**89.6**	96.0	**93.10**	**91.8**

connections and asymmetric convolutions, achieving a 78.8% of OA. The lightweight SCAttNet rebuilds the attention steps for feature refinement in remote sensing images, resulting in about a 1% increase of OA over DANet and CBAM. HAMNet incorporates more contextual information for higher accuracy. LANet produces the second-best OA by attentively bridging the gap between high- and low-level features. ResUNet-a achieves similar OA as LANet by hybridizing abundant optimization strategies, and however, LANet has lower complexity and fewer parameters than ResUNet-a.

Fig. 10 visualizes the prediction outcome on the testing set. Attention-embedded networks yield better quality than earlier encoder–decoder architecture. Moreover, SAPNet segments EODs and WSOs with the best consistency with the ground truth. The fine boundary and accurate localization and classification of all classes are realized with a simple strategy. To sum up, the WDOs and WSOs in the Vaihingen dataset are well-distinguished by SAPNet, due to unbiased contextual affinity. This affinity jointly models the spatial and channel information that boosts the discriminative capability of semantic segmentation models effectively.

2) Results on the ISPRS Potsdam Dataset: As evident from Table IV, attention-based methods outperform baseline encoder–decoder methods. CBAM and DANet still express the scalability of the attention mechanism in capturing nonlocal correlations, bringing the OA to almost 88%. However, the heterogeneity of remote sensing images makes natural image-specific approaches far from optimal. With spatial and channel context information, HMANet produces the second-highest performance on the ISPRS Potsdam dataset, even slightly higher than the complicated ResUNet-a. Due to the injection of sufficient contextual affinity, SAPNet achieves the highest OA of 91.8% and AF of 93.10.

Visual inspections are shown in Fig. 11. We can clearly see that SAPNet segments both WDOs and WSOs much finer than other SOTA methods. Standard encoder–decoder variants are generally inferior in mapping ground objects. The WDOs are incomplete. Moreover, the inconsistent blurs are mapped inside the WSOs. With the incorporation of multiscale context, DeepLab V3+ partially resolves the blurs. While injecting attentive context, the main parts and objects are well-distinguished, as CBAM and DANet work. However, attention bias makes this kind of method suboptimal. After synergistically modeling contextual affinity, the spatial details and spectral correlations are well-extracted and integrated. As shown in Fig. 11(b), the impervious surfaces are consistently delineated as the ground truth. The buildings on the left are well-outlined too. As a specific category of WDOs, cars are error-prone by comparative methods with blurry edges in Fig. 11(a). Likewise, the compared methods

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 20:07:07 UTC from IEEE Xplore. Restrictions apply.

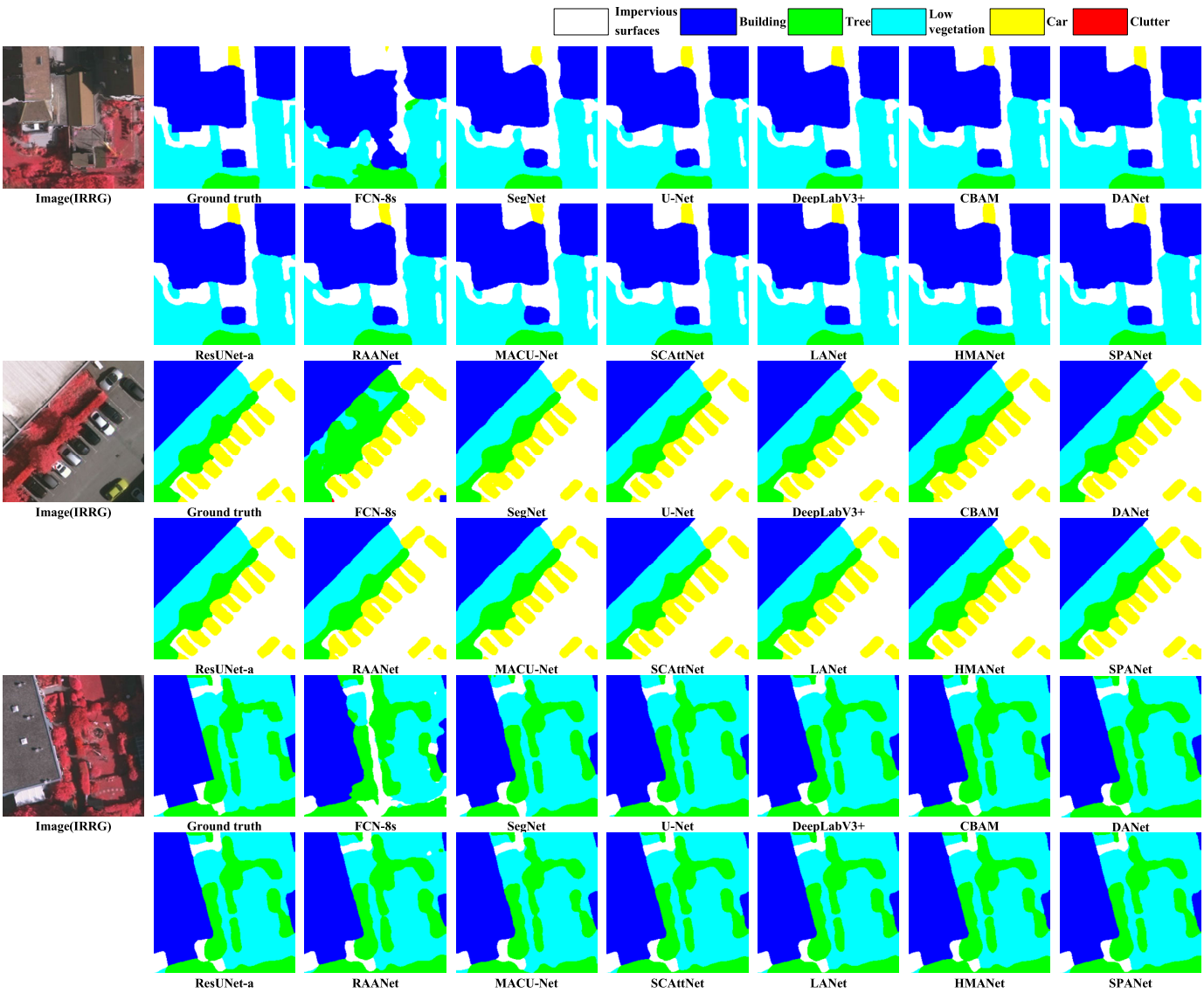


Fig. 10. Visual results on the ISPRS Vaihingen testing set (randomly selected samples).

TABLE V

RESULTS ON THE DEEPGLOBE DATASET. CLASSWISE F1-SCORE, AF, AND OA ARE LISTED, WHERE THE BOLD TEXT INDICATES THE BEST RESULTS

Methods	Urban land	Agriculture land	Forest land	Water	Barren land	Rangeland	AF	OA
FCN-8s [15]	82.1	84.0	63.3	81.5	86.7	72.7	78.40	79.4
SegNet [16]	84.2	84.6	60.0	84.2	91.5	79.1	80.62	82.4
U-Net [17]	82.6	90.0	68.0	88.5	94.5	85.5	84.85	85.9
DeepLabV3+ [18]	83.1	86.5	71.6	88.0	94.1	90.6	85.64	87.1
CBAM [33]	84.9	92.3	73.4	88.8	93.9	92.4	87.59	89.8
DANet [32]	87.1	92.1	75.2	90.1	94.4	94.0	88.83	90.7
ResUNet-a [58]	92.5	95.1	82.9	87.6	95.3	94.3	91.30	91.8
RAANet [66]	87.2	91.9	76.7	85.3	93.2	91.8	87.68	89.9
MACU-Net [67]	87.8	92.3	78.3	86.6	92.9	92.5	88.39	90.2
SCAttNet [37]	88.2	92.2	74.5	90.3	96.6	95.2	89.49	90.6
LANet [64]	87.3	88.9	75.9	88.9	94.3	90.6	87.64	89.6
HMANet [38]	92.2	94.9	82.7	87.4	95.1	94.1	91.06	91.5
SAPNet (ours)	**93.6**	**96.0**	**85.1**	**91.8**	**96.8**	**95.2**	**93.09**	**93.1**

are unable to handle complex scenes that cover multiple contiguous classes. The visual inspections in Fig. 11(c) exhibit high consistency with ground truth by SAPNet. With the joint-modeled affinity, SAPNet strengthens the discriminability of learned representations of cars, leading to better visual performance.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 20:07:07 UTC from IEEE Xplore. Restrictions apply.

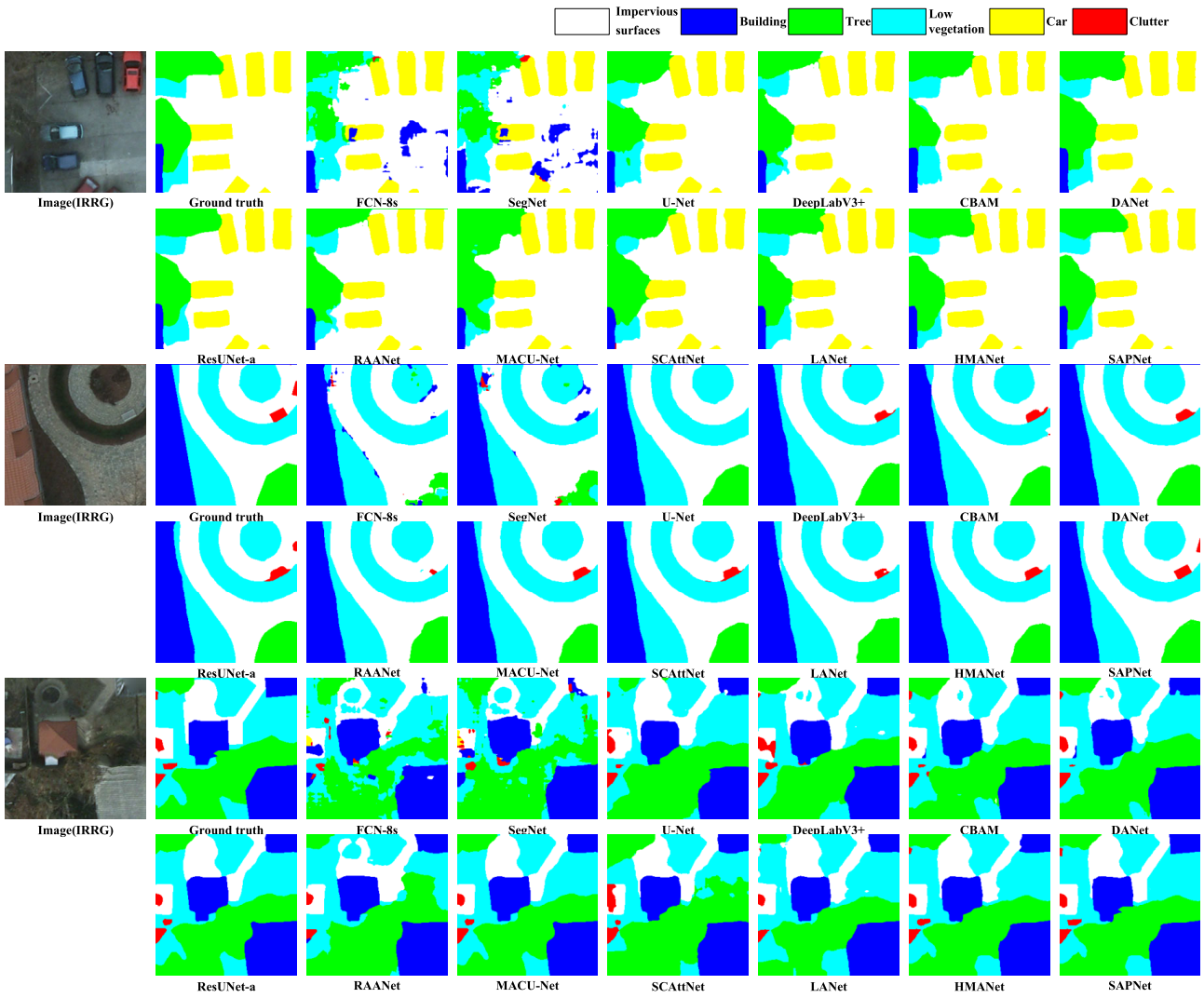


Fig. 11. Visual results on the ISPRS Potsdam testing set (randomly selected samples).

In summary, the quantitative and qualitative evaluations generally agree with the effects of contextual affinity. Moreover, the synergistic attention makes SAPNet extract comprehensive context at multiple scales, giving significant accuracy improvement on the ISPRS Potsdam dataset.

3) Results on the DeepGlobe Dataset: The results on DeepGlobe satellite images are shown in Table V. Although the spatial resolution is lower than the aerial images and the observation range is wider than ISPRS benchmarks, the well-annotated masks and massive data provide the preferable preconditions for training the optimal network. Statistically, 92.75 AF and 92.9% OA are attained by SAPNet. Compared with the state-of-the-art remote sensing specific methods, the AF and OA have increased by 1.79% and 1.3% compared to ResUNet-a, respectively. In addition to overall evaluation, categorywise F1-score is also in the lead. Moreover, a similar trend with ISPRS benchmarks is marked. Attentively modeling contextual affinity helps the network to learn more discriminative representations to boost the dense predictions. CBAM sequentially incorporates spatial attention module and CAM, increasing AF and OA by 1.95 and 2.7% than DeepLab V3+, respectively. Moreover, remote sensing-specific methods specify the essence of ground objects and design corresponding fine-tuning tricks or transformations. As a result, these methods produce better performance. Specifically, ResUNet-a exhibits the second-best OA with 91.8%. In summary, incorporating spatial correlations into channelwise contextual affinity also lends strong support to recognizing ground objects from satellites.

Visual inspections are shown in Fig. 12. Three samples are selected from the testing set. WDOs and WSOs should be segmented with high certainty and consistency by a robust neural network. In Fig. 12(a), the rangeland and the neighboring agricultural land share similar visual features and irregular shapes. It is thus hard to be classified. ResUNet-a integrates multiple strategies that attain a comparable smooth mask. Three attention variants of remote sensing-specific methods also boost the results significantly. However, the attention bias produces blurs inside rangeland and coarse-grained boundaries between different objects. Since the water areas are characterized by strong spectral absorbance in visible and NIR bands compared to other kinds of ground objects, which is vulnerable to the

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 20:07:07 UTC from IEEE Xplore. Restrictions apply.

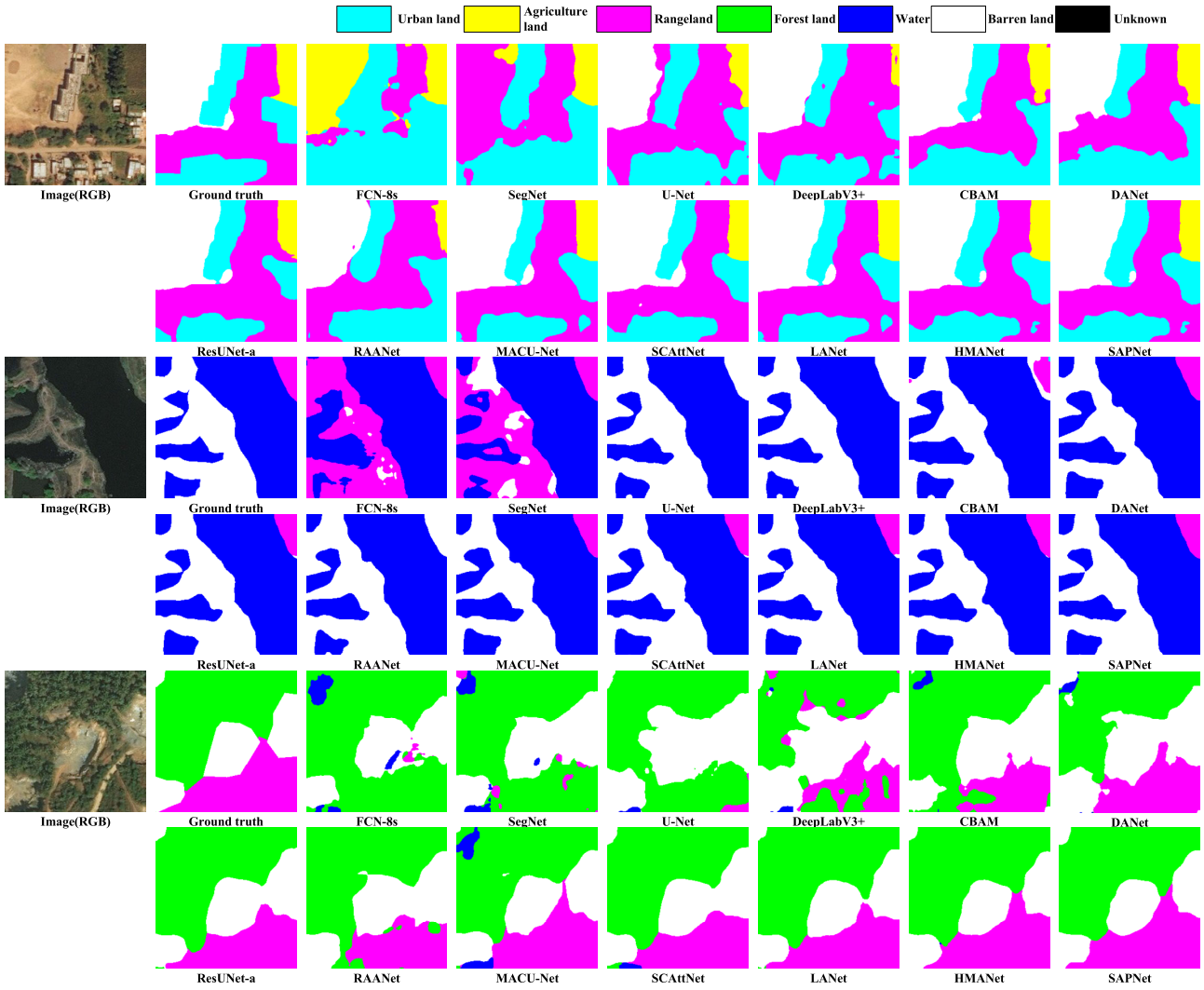


Fig. 12. Visual results on the DeepGlobe testing set (randomly selected samples).

surrounding context, accurate water extraction requires discriminative features. Besides channelwise correlations, spatial details are also necessary because water areas always present irregular shapes and local connectivity. SAPNet turns these requirements into reality. Various water areas with different shapes and surfaces are well-mapped. To sum up, SAPNet explicitly expresses the best qualitative visualization due to the synergistically modeled contextual affinity at multiple scales.

TABLE VI

EFFICIENCY EVALUATION OF BENCHMARK MODELS. INFERENCE TIME IS CALCULATED AS AN AVERAGE TIME FOR 500 EXPERIMENTS

Methods	Inference time (ms)	FLOPs (G)
PAM-only	22.57	65.32
CAM-only	10.26	16.26
CAM+PAM sequential	33.61	70.69
CAM+PAM parallel	29.12	70.69
SAP (with proposed SAM)	13.79	21.36

F. Efficiency Analysis

The inference time and floating-point operations (FLOPs) of different attention models embedded into SAPNet are presented in Table VI. The FLOPs are calculated with the input image size of $256 \times 256 \times 3$. The FLOPs of the proposed SAP are slightly higher than those of CAM-only. However, the FLOPs of SAP are far less than those of PAM-only, CAM + PAM sequential, and CAM + PAM parallel, which involve more than three times the FLOPs of SAP. A significant reduction of FLOPs is by more than 69% in producing channel and spatial attention maps with sequential or parallel models. As to time consumption, the inference time is an average of inferencing 500 subpatches for all methods. SAP ranks in the second order (see details in Table VI). A margin of 3.53 ms is reported compared to PAM-only in exchange for an increase of 5.1% in OA.

In conclusion, the synergistic modeled contextual affinity benefits the network with richer clues for inference. These clues boost the discriminability for learned representations. Specifically, the efficiency evaluation shows a promising inference speed and low FLOPs requirement compared with PAM-only, CAM + PAM sequential, and CAM + PAM parallel models.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 20:07:07 UTC from IEEE Xplore. Restrictions apply.

V. Conclusion

In this study, we first specify the attention bias of existing paradigms on segmenting ground objects in remote sensing images. Instead of separately modeling spatial and channel attention maps or with the postfusion operation, the proposed SAM jointly models spatialwise and channelwise correlations in one attention map, which is informative of spatial and channel interactions. Incorporating the SAM, the developed SAPNet takes care of the multiscale inherence, reaching competitive results on three representative benchmarks.

In the future, two aspects will be further studied. On the one hand, the feature representation capacity of latent geometric property in Euclidean space is limited. It is promising to form a non-Euclidean feature space that enables high-fidelity feature representation. On the other hand, a few-shot semantic segmentation network for remote sensing images is required due to the high costs of labeling samples.

References

[1] Q. Liu, M. Kampffmeyer, R. Jenssen, and A.-B. Salberg, "Dense dilated convolutions' merging network for land cover classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 9, pp. 6309–6320, Sep. 2020.

[2] X. Yuan, J. Shi, and L. Gu, "A review of deep learning methods for semantic segmentation of remote sensing imagery," *Expert Syst. Appl.*, vol. 169, May 2021, Art. no. 114417.

[3] C. Zhang, W. Jiang, Y. Zhang, W. Wang, Q. Zhao, and C. Wang, "Transformer and CNN hybrid deep neural network for semantic segmentation of very-high-resolution remote sensing imagery," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 4408820.

[4] S. Du, S. Du, B. Liu, and X. Zhang, "Mapping large-scale and fine-grained urban functional zones from VHR images using a multi-scale semantic segmentation network and object based approach," *Remote Sens. Environ.*, vol. 261, Aug. 2021, Art. no. 112480.

[5] S. Scepanovic, O. Antropov, P. Laurila, Y. Rauste, V. Ignatenko, and J. Praks, "Wide-area land cover mapping with Sentinel-1 imagery using deep learning semantic segmentation models," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 14, pp. 10357–10374, 2021.

[6] B. Cui, W. Jing, L. Huang, Z. Li, and Y. Lu, "SANet: A sea–land segmentation network via adaptive multiscale feature learning," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 14, pp. 116–126, 2021.

[7] H.-F. Zhong, H.-M. Sun, D.-N. Han, Z.-H. Li, and R.-S. Jia, "Lake water body extraction of optical remote sensing images based on semantic segmentation," *Int. J. Speech Technol.*, vol. 52, no. 15, pp. 17974–17989, Dec. 2022.

[8] Q. Xu, C. Ouyang, T. Jiang, X. Yuan, X. Fan, and D. Cheng, "MFFENet and ADANet: A robust deep transfer learning method and its application in high precision and fast cross-scene recognition of earthquake-induced landslides," *Landslides*, vol. 19, no. 7, pp. 1617–1647, Jul. 2022.

[9] B. Du et al., "Landslide susceptibility prediction based on image semantic segmentation," *Comput. Geosci.*, vol. 155, Oct. 2021, Art. no. 104860.

[10] M. Nikzad, Y. Gao, and J. Zhou, "An attention-based lattice network for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5526215.

[11] A. Farooq, X. Jia, J. Hu, and J. Zhou, "Transferable convolutional neural network for weed mapping with multisensor imagery," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 4404816.

[12] Z. Han, D. Hong, L. Gao, J. Yao, B. Zhang, and J. Chanussot, "Multimodal hyperspectral unmixing: Insights from attention networks," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5524913.

[13] K. Yan, M. Zhou, L. Liu, C. Xie, and D. Hong, "When pansharpening meets graph convolution network and knowledge distillation," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5408915.

[14] D. Hong, J. Yao, D. Meng, Z. Xu, and J. Chanussot, "Multimodal GANs: Toward crossmodal hyperspectral–multispectral image segmentation," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 6, pp. 5103–5113, Jun. 2021.

[15] J. Long, E. Shelhamer, and T. Darrell, "Fully convolutional networks for semantic segmentation," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2015, pp. 3431–3440.

[16] V. Badrinarayanan, A. Kendall, and R. Cipolla, "SegNet: A deep convolutional encoder–decoder architecture for image segmentation," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 39, no. 12, pp. 2481–2495, Dec. 2017.

[17] O. Ronneberger, P. Fischer, and T. Brox, "U-Net: Convolutional networks for biomedical image segmentation," in *Proc. Int. Conf. Med. Image Comput. Comput.-Assist. Intervent.*, 2015, pp. 234–241.

[18] L. C. Chen, G. Papandreou, and I. Kokkinos, "DeepLab: Semantic image segmentation with deep convolutional nets, atrous convolution, and fully connected CRFs," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 40, no. 4, pp. 834–848, Jun. 2016.

[19] Z. Li et al., "Cascaded multiscale structure with self-smoothing atrous convolution for semantic segmentation," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5605713.

[20] T. Panboonyuen, K. Jitkajornwanich, S. Lawawirojwong, P. Srestasathiern, and P. Vateekul, "Semantic labeling in remote sensing corpora using feature fusion-based enhanced global convolutional network with high-resolution representations and depthwise atrous convolution," *Remote Sens.*, vol. 12, no. 8, p. 1233, Apr. 2020.

[21] L. Gao et al., "STransFuse: Fusing swin transformer and convolutional neural network for remote sensing image semantic segmentation," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 14, pp. 10990–11003, 2021.

[22] H. Guo, B. Du, L. Zhang, and X. Su, "A coarse-to-fine boundary refinement network for building footprint extraction from remote sensing imagery," *ISPRS J. Photogramm. Remote Sens.*, vol. 183, pp. 240–252, Jan. 2022.

[23] Y. Wang, W. Yu, and Z. Fang, "Multiple kernel-based SVM classification of hyperspectral images by combining spectral, spatial, and semantic information," *Remote Sens.*, vol. 12, no. 1, p. 120, Jan. 2020.

[24] S. Ghaderizadeh, D. Abbasi-Moghadam, A. Sharifi, N. Zhao, and A. Tariq, "Hyperspectral image classification using a hybrid 3D-2D convolutional neural networks," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 14, pp. 7570–7588, 2021.

[25] X. Tang et al., "Hyperspectral image classification based on 3-D octave convolution with spatial–spectral attention network," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 3, pp. 2430–2447, Mar. 2021.

[26] X. Wang and Z. Liang, "Hybrid network model based on 3D convolutional neural network and scalable graph convolutional network for hyperspectral image classification," *IET Image Process.*, vol. 17, no. 1, pp. 256–273, Jan. 2023.

[27] X. Li, L. Lei, and G. Kuang, "Multilevel adaptive-scale context aggregating network for semantic segmentation in high-resolution remote sensing images," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, pp. 1–5, 2022.

[28] D. Zhang et al., "Edge-aware and spectral–spatial information aggregation network for multispectral image semantic segmentation," *Eng. Appl. Artif. Intell.*, vol. 114, Sep. 2022, Art. no. 105070.

[29] A. Vaswani et al., "Attention is all you need," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 30, 2017, pp. 5998–6008.

[30] J. Hu, L. Shen, S. Albanie, G. Sun, and E. Wu, "Squeeze-and-excitation networks," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 42, no. 8, pp. 2011–2023, Apr. 2020.

[31] X. Wang, R. Girshick, A. Gupta, and K. He, "Non-local neural networks," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2018, pp. 7794–7803.

[32] J. Fu et al., "Dual attention network for scene segmentation," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2019, pp. 3141–3149.

[33] S. Woo, J. Park, J.-Y. Lee, and I. S. Kweon, "CBAM: Convolutional block attention module," in *Proc. Eur. Conf. Comput. Vis.*, 2018, pp. 3–19.

[34] Y. Yuan, X. Chen, and J. Wang, "Object-contextual representations for semantic segmentation," in *Proc. Eur. Conf. Comput. Vis.*, 2020, pp. 173–190.

[35] X. X. Zhu et al., "Deep learning in remote sensing: A comprehensive review and list of resources," *IEEE Geosci. Remote Sens. Mag.*, vol. 5, no. 4, pp. 8–36, Dec. 2017.

[36] L. Zhang and L. Zhang, "Artificial intelligence for remote sensing data analysis: A review of challenges and opportunities," *IEEE Geosci. Remote Sens. Mag.*, vol. 10, no. 2, pp. 270–294, Jun. 2022.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 20:07:07 UTC from IEEE Xplore. Restrictions apply.

[37] H. Li, K. Qiu, L. Chen, X. Mei, L. Hong, and C. Tao, "SCAttNet: Semantic segmentation network with spatial and channel attention mechanism for high-resolution remote sensing images," *IEEE Geosci. Remote Sens. Lett.*, vol. 18, no. 5, pp. 905–909, May 2020.

[38] R. Niu, X. Sun, Y. Tian, W. Diao, K. Chen, and K. Fu, "Hybrid multiple attention network for semantic segmentation in aerial images," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5603018.

[39] F. Babiloni, I. Marras, G. Slabaugh, and S. Zafeiriou, "TESA: Tensor element self-attention via matricization," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2020, pp. 13942–13951.

[40] *ISPRS 2D Semantic Labeling—Vaihingen Data*. Accessed: Dec. 22, 2021. [Online]. Available: https://www.isprs.org/education/benchmarks/UrbanSemLab/2d-sem-label-vaihingen.aspx

[41] *ISPRS 2D Semantic Labeling Contest—Potsdam*. Accessed: Dec. 22, 2021. [Online]. Available: https://www.isprs.org/education/benchmarks/UrbanSemLab/2d-sem-label-potsdam.aspx

[42] I. Demir et al., "DeepGlobe 2018: A challenge to parse the earth through satellite images," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR) Workshops*, Jun. 2018, pp. 172–181.

[43] G. Brauwers and F. Frasincar, "A general survey on attention mechanisms in deep learning," *IEEE Trans. Knowl. Data Eng.*, early access, Nov. 9, 2021, doi: 10.1109/TKDE.2021.3126456.

[44] Z. Li, F. Liu, W. Yang, S. Peng, and J. Zhou, "A survey of convolutional neural networks: Analysis, applications, and prospects," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 33, no. 12, pp. 6999–7019, Dec. 2022.

[45] M. Yang, K. Yu, C. Zhang, Z. Li, and K. Yang, "DenseASPP for semantic segmentation in street scenes," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2018, pp. 3684–3692.

[46] M. M. Bronstein, J. Bruna, Y. LeCun, A. Szlam, and P. Vandergheynst, "Geometric deep learning: Going beyond Euclidean data," *IEEE Signal Process. Mag.*, vol. 34, no. 4, pp. 18–42, Jul. 2017.

[47] Z.-Q. Liu, P. Tang, W. Zhang, and Z. Zhang, "CNN-enhanced heterogeneous graph convolutional network: Inferring land use from land cover with a case study of park segmentation," *Remote Sens.*, vol. 14, no. 19, p. 5027, Oct. 2022.

[48] X. Tang, M. Li, J. Ma, X. Zhang, F. Liu, and L. Jiao, "EMTCAL: Efficient multiscale transformer and cross-level attention learning for remote sensing scene classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5626915.

[49] M. Gong, J. Li, Y. Zhang, Y. Wu, and M. Zhang, "Two-path aggregation attention network with quad-patch data augmentation for few-shot scene classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 4511616.

[50] L. Li, X. Yao, G. Cheng, M. Xu, J. Han, and J. Han, "Solo-to-collaborative dual-attention network for one-shot object detection in remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5607811.

[51] Y. Liu, Q. Li, Y. Yuan, Q. Du, and Q. Wang, "ABNet: Adaptive balanced network for multiscale object detection in remote sensing imagery," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5614914.

[52] W. Zhang, J. Li, and Z. Hua, "Attention-based multistage fusion network for remote sensing image pansharpening," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5405416.

[53] D. Lei, P. Chen, L. Zhang, and W. Li, "MCANet: A multidimensional channel attention residual neural network for pansharpening," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5411916.

[54] Z. Lv, F. Wang, G. Cui, J. A. Benediktsson, T. Lei, and W. Sun, "Spatial–spectral attention network guided with change magnitude image for land cover change detection using remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 4412712.

[55] Q. Shi, M. Liu, S. Li, X. Liu, F. Wang, and L. Zhang, "A deeply supervised attention metric-based network and an open aerial image dataset for remote sensing change detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5604816.

[56] L. Ding et al., "Looking outside the window: Wide-context transformer for the semantic segmentation of high-resolution remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 4410313.

[57] K. Yue, L. Yang, R. Li, W. Hu, F. Zhang, and W. Li, "TreeUNet: Adaptive tree convolutional neural networks for subdecimeter aerial image segmentation," *ISPRS J. Photogramm. Remote Sens.*, vol. 156, pp. 1–13, Oct. 2019.

[58] F. I. Diakogiannis, F. Waldner, P. Caccetta, and C. Wu, "ResUNet—A: A deep learning framework for semantic segmentation of remotely sensed data," *ISPRS J. Photogramm. Remote Sens.*, vol. 162, pp. 94–114, Apr. 2020.

[59] X. Li et al., "Dual attention deep fusion semantic segmentation networks of large-scale satellite remote-sensing images," *Int. J. Remote Sens.*, vol. 42, no. 9, pp. 3583–3610, May 2021.

[60] J. Huang, X. Zhang, Y. Sun, and Q. Xin, "Attention-guided label refinement network for semantic segmentation of very high resolution aerial orthoimages," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 14, pp. 4490–4503, 2021.

[61] X. Yang et al., "An attention-fused network for semantic segmentation of very-high-resolution remote sensing imagery," *ISPRS J. Photogramm. Remote Sens.*, vol. 177, pp. 238–262, Jul. 2021.

[62] X. Li, F. Xu, R. Xia, X. Lyu, H. Gao, and Y. Tong, "Hybridizing cross-level contextual and attentive representations for remote sensing imagery semantic segmentation," *Remote Sens.*, vol. 13, no. 15, p. 2986, Jul. 2021.

[63] R. Liu, L. Mi, and Z. Chen, "AFNet: Adaptive fusion network for remote sensing image semantic segmentation," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 9, pp. 7871–7886, Sep. 2021.

[64] L. Ding, H. Tang, and L. Bruzzone, "LANet: Local attention embedding to improve the semantic segmentation of remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 1, pp. 426–435, Jan. 2021.

[65] K. Han et al., "A survey on vision transformer," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 45, no. 1, pp. 87–110, Jan. 2023.

[66] R. Liu et al., "RAANet: A residual ASPP with attention framework for semantic segmentation of high-resolution remote sensing images," *Remote Sens.*, vol. 14, no. 13, p. 3109, 2022.

[67] R. Li, C. Duan, S. Zheng, C. Zhang, and P. M. Atkinson, "MACU-Net for semantic segmentation of fine-resolution remotely sensed images," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, pp. 1–5, 2022.



Xin Li received the B.S. degree from Zhengzhou University, Zhengzhou, China, in 2013, and the M.S. and Ph.D. degrees from Hohai University, Nanjing, China, in 2017 and 2021, respectively.

He is currently a Post-Doctoral Researcher with Hohai University. His research interests include computer vision and remote sensing image processing.



Feng Xu is currently a Professor with the College of Computer and Information, Hohai University, Nanjing, China. He has published more than 100 papers in refereed international journals and conference proceedings. His research interests include smart water, remote sensing applications, cloud computing, and domain software engineering.



Fan Liu (Member, IEEE) received the B.S. and Ph.D. degrees from the Nanjing University of Science and Technology, Nanjing, China, in 2009 and 2015, respectively.

In 2008, he was a Researcher with Ajou University, Suwon, South Korea. In 2014, he was with Microsoft Research Asia, Beijing, China. He is currently a Professor with Hohai University, Nanjing. His research interests include computer vision, pattern recognition, and machine learning.

Dr. Liu also serves as a Reviewer for the IEEE TRANSACTIONS ON NEURAL NETWORKS AND LEARNING SYSTEMS, the IEEE TRANSACTIONS ON KNOWLEDGE AND DATA ENGINEERING, the *ACM Transactions on Intelligent Systems and Technology*, *Information Sciences*, *Neurocomputing*, and *Pattern Analysis and Applications*.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 20:07:07 UTC from IEEE Xplore. Restrictions apply.



Xin Lyu received the Ph.D. degree from Hohai University, Nanjing, China, in 2013.

He has published over 20 papers in refereed international journals and conference proceedings. His research interests include cryptography, network information security, and privacy-preserving theory and technology.



Yao Tong received the B.S. degree in electronic information engineering and the M.S. degree in software engineering from Hohai University, Nanjing, China, in 2015 and 2017, respectively, and the Ph.D. degree from Zhengzhou University, Zhengzhou, China, in 2022.

She is currently a Lecturer with the School of Artificial Intelligence and Information Technology, Nanjing University of Chinese Medicine. She has been a Visiting Student with the Department of Biomedical Informatics and Medical Education, University of Washington, Seattle, WA, USA. Her research interests include machine learning, computer vision, database systems, and clinical informatics.



Zhennan Xu received the M.S. degree from Hohai University, Nanjing, China, in 2021, where he is currently pursuing the Ph.D. degree.

His research interests include pansharpening and remote sensing image semantic segmentation.



Jun Zhou (Senior Member, IEEE) received the B.S. degree in computer science and the B.E. degree in international business from the Nanjing University of Science and Technology, Nanjing, China, in 1996 and 1998, respectively, the M.S. degree in computer science from Concordia University, Montreal, QC, Canada, in 2002, and the Ph.D. degree in computing science from the University of Alberta, Edmonton, AB, Canada, in 2006.

He was a Research Fellow with the Research School of Computer Science, The Australian National University, Canberra, ACT, Australia, and a Researcher with the Canberra Research Laboratory, National Information and Communications Technology Australia, Canberra. In 2012, he joined the School of Information and Communication Technology, Griffith University, Nathan, QLD, Australia, where he is currently an Associate Professor. His research interests include pattern recognition, computer vision, and spectral imaging and their applications in remote sensing and environmental informatics.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 20:07:07 UTC from IEEE Xplore. Restrictions apply.